

Yashveer Jain

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Software Developer and Machine Learning Engineer with a Master's in Robotics from the University of Maryland. Over 2 years of experience in developing and deploying machine learning models, orchestrating ROS-based autonomous control systems, and working with embedded AI systems. Proficient in Python, C++, and various machine learning libraries.

SKILLS AND LANGUAGES

Machine Learning: TensorFlow, PyTorch, Pandas, Matplotlib, OpenCV, Scikit-learn, Vector-c++, Huggingface.

Programming Languages: Python, C, C++

Development Tools: Git, Docker, Agile Development

Cloud Platforms: AWS (Certified Cloud Practitioner), Azure

Robotics: ROS (Robot Operating System), ROS2, Arduino, Raspberry Pi, Nvidia Jetson.

WORK EXPERIENCE

Embedded AI Intern - Renesas Electronics, USA

May 2023–Aug 2023

- **Trained and deployed SVM and logistic regression models on Renesas RA6 Series and RL78, achieving 95% accuracy for an Asset Tracking Project**, enabling efficient asset management.
- **Ensured success of edge computing projects** by developing and implementing machine learning models optimized for resource-constrained embedded systems.

Machine learning Engineer - AIMonk Labs, India

Aug 2020–Aug 2022

- **Developed and deployed a robust testing infrastructure for AI models**, integrating FastAPI backend on AWS EC2 instance, streamlining the model development lifecycle.
- **Optimized OCR algorithms using Transformers implemented on PyTorch, achieving a 60% accuracy boost**, significantly improving text recognition capabilities.
- **Enhanced car detection AI by 20% using DeepLabv3 and YOLOv7 integration**, leading to more accurate object detection in autonomous systems.
- **Collaborated with cross-functional teams on Azure DevOps for model deployment on AWS SageMaker**, ensuring seamless integration and deployment of machine learning models.

Research Intern - Swaayatt Robotics, India

Jan 2020–May 2020

- Implemented **Kalman and particle filters with LK optical flow using OpenCV** for vehicle and pedestrian tracking in autonomous driving systems.
- Implemented UnDeepVO paper unsupervised deep learning visual odometry paper using **TensorFlow 2.0 and Keras** for depth estimation and translation and rotation motion, **increasing the SLAM accuracy by 20%**.

RESEARCH AND TEACHING EXPERIENCE

Grader, University of Maryland (Spring 2024)

Jan 2024–May 2024

ENPM808z – Cognitive Robotics with Prof. Yantian Zha, a graduate course at UMD. Adept at assessing student progress in integrating human cognition with robotics and AI, using simulators like **NVIDIA Isaac Sim**, and **Meta Habitat 2.0**.

Graduate Teaching Assistant, University of Maryland (Fall 2023)

Aug 2023–Dec 2023

ENPM667 – Control of Robotic Systems with Prof. Waseem Malik, a graduate course at UMD. Providing targeted tutoring in **Linearization techniques** and control system topics like **PID, LQR, LQG**.

Graduate Research Assistant, University of Maryland (Fall 2022–Spring 2023)

Aug 2022–May 2023

TinyDepth: Generalized Neural Metric Depth for Palm-sized Robots. Under Review Nature npj Robotics.

Designed sub-6cm microdrone with embedded Raspberry Pi and ROS-based vision pipeline for autonomous planning and control (Perception & Robotics Group, Prof. Yiannis Aloimonos).

PROJECTS

Irona - [GitHub](#)

Aug 2022–Dec 2022

Designed a service robot **using ROS, computer vision, and C++** for object manipulation, from one place to another.

Home Automation - [GitHub](#)

Jun 2021–Aug 2021

Established a home server for device connectivity **using FastAPI, MQTT, Arduino, Raspberry Pi**. Demonstrating proficiency in hardware integration and IoT systems.

Third eye - [GitHub](#)

Oct 2019–Dec 2019

Built blind navigation device using **Siamese Network** (face recognition), **Yolo9000** (object detection), and **Text2Speech**.

Search and Rescue Drone - [GitHub](#)

Jul 2019–Oct 2019

Developed an AI Drone to detect human beings using **RentinaNet** for object detection in hazardous areas and send their locations to the Rescue team, showcasing practical application of AI in real-world scenarios.

EDUCATION

Master of Engineering (Robotics) (GPA:3.85/4)

Aug 2022–May 2024

University of Maryland – College Park (UMD), USA

Bachelor of Engineering (Mechatronics) (GPA:9.44/10)

July 2016– July 2020

Manipal University Jaipur, Rajasthan, India

CERTIFICATE

Building Transformer-Based Natural Language Processing Applications (Nvidia)

Feb 2024

Training and Deploying **BERT Transformer Model** on **NER and Text classification** and deployed on **Triton Server**.

AWS Certified Cloud Practitioner (AWS)

Mar 2024